CURVE-DEFORMED CONSTANT P-MEAN-CURVATURE SURFACES IN THE HEISENBERG GROUP AND THEIR EXPLICIT NORMAL-COORDINATE INVARIANTS

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Allowed standard prerequisites (editorial).

- Research prerequisites named in the original article. Full original bibliography retained; all source-local definitions and supporting arguments retained.

1. Introduction

This article is an extension of the previous paper [5], in which we studied the constant p-mean curvature surfaces in the Heisenberg group H_1. In [5], we focused on the foundation of the theory and paid more attention to the investigation of p-minimal surfaces. However, in the present article, instead of theory, we mainly focus on the examples, including an approach to construct constant p-mean curvature surfaces.

Recall that the Heisenberg group H_1 is the space $\mathbb{R}^3$ with the associated group multiplication

$$(x_1, y_1, z_1) \circ (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2 + y_1 x_2 - x_1 y_2),$$

which is a 3-dimensional Lie group. The space of all left-invariant vector fields is spanned by the following three vector fields:

$$\mathring{e}_1 = \frac{\partial}{\partial x} + y\frac{\partial}{\partial z}, \qquad \mathring{e}_2 = \frac{\partial}{\partial y} - x\frac{\partial}{\partial z} \qquad \text{and} \qquad T = \frac{\partial}{\partial z}.$$

The Heisenberg dilation (scaling) by the factor $\delta > 0$ is the map $D_\delta \colon H_1 \to H_1$ defined by $D_\delta(x, y, z) = \left(\delta x, \delta y, \delta^2 z\right)$ for any $(x, y, z) \in H_1$ (see [6]).

The standard contact bundle on H_1 is the subbundle ξ of the tangent bundle TH_1 spanned by $\mathring{e}_1$ and $\mathring{e}_2$. It is also defined to be the kernel of the contact form $\Theta = \mathrm{d}z + x\mathrm{d}y - y\mathrm{d}x$. The CR structure on H_1 is the endomorphism $J \colon \xi \to \xi$ defined by $J(\mathring{e}_1) = \mathring{e}_2$ and $J(\mathring{e}_2) = -\mathring{e}_1$. One can view H_1 as a pseudo-hermitian manifold with (J, Θ) as the standard pseudo-hermitian structure. There is a naturally associated connection ∇ if we regard all these left-invariant vector fields $\mathring{e}_1$, $\mathring{e}_2$, and T as parallel vector fields. A naturally associated metric on H_1 is the adapted metric g_Θ, which is defined by $g_\Theta = \mathrm{d}\Theta(\cdot, J\cdot) + \Theta^2$. It is equivalent to defining the metric regarding $\mathring{e}_1$, $\mathring{e}_2$, and T as an orthonormal frame field. We sometimes use $\langle \cdot, \cdot \rangle$ to denote the adapted metric. In this paper, we use the adapted metric to measure the lengths, angles of vectors, and so on.

Suppose Σ is a surface in the Heisenberg group H_1. There is a one-form I on Σ induced from the adapted metric g_Θ. This induced metric is defined on the whole surface Σ and is called the first fundamental form of Σ. The intersection $T\Sigma \cap \xi$ is integrated to be a singular foliation on Σ called the characteristic foliation. Each leaf is called a characteristic curve. A point $p \in \Sigma$ is called a singular point if the tangent plane $T_p\Sigma$ coincides with the contact plane ξ_p; otherwise, p is called a regular (or non-singular) point. Generically, a point $p \in \Sigma$ is a regular point, and the set of all regular points is called the regular part of Σ. In this paper, we always assume that the surface Σ is of class C^2, but of class C^∞ on the regular part. On the regular part, we can choose a unit vector field e_1 such that e_1 defines the characteristic foliation. The vector e_1 is determined up to a sign. Let $e_2 = Je_1$. Then $\{e_1, e_2\}$ forms an orthonormal frame field of the contact bundle ξ. We usually call the vector field e_2 a horizontal normal vector field. Then the p-mean curvature H of the surface Σ is defined by $\nabla_{e_1} e_2 = -He_1$. The p-mean curvature H is only defined on the regular part of Σ. If $H = c$, which is a constant on the whole regular part, we call the surface a constant p-mean curvature surface. In particular, if $c = 0$, it is a p-minimal surface. There also exists a function α defined on the regular part such that $\alpha e_2 + T$ is tangent to the surface Σ. We call this function the α-function of Σ. It is uniquely determined up to a sign, which depends on the choice of the characteristic direction e_1. Define $\hat{e}_1 = e_1$ and $\hat{e}_2 = \frac{\alpha e_2 + T}{\sqrt{1+\alpha^2}}$, then $\{\hat{e}_1, \hat{e}_2\}$ forms an orthonormal frame field of the tangent bundle $T\Sigma$. Notice that $\hat{e}_2$ is uniquely determined and independent of the choice of the characteristic direction e_1. In [3, 4], it was shown that these three invariants, I, e_1, and α, form a complete set of invariants for constant p-mean curvature surfaces with $H = c$ in H_1. Namely, for any two surfaces with the same constant p-mean curvature having the same I, α, e_1, they are differed only by a Heisenberg symmetry. In particular, if $\Sigma \subset H_1$ is a *constant* p-mean curvature surface with $H = c$, then in

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terms of a compatible coordinate system $(U; x, y)$, which means $e_1 = \frac{\partial}{\partial x}$, the integrability condition (see [5]) is reduced to

$$-a_x + a\frac{b_x}{b} = \frac{c\alpha}{\left(1+\alpha^2\right)^{1/2}}, \qquad -\frac{b_x}{b} = 2\alpha + \frac{\alpha\alpha_x}{1+\alpha^2},$$

$$\alpha_{xx} + 6\alpha\alpha_x + 4\alpha^3 + c^2\alpha = 0, \tag{1.1}$$

where the two functions a and b are a representation of the first fundamental form I in the following sense that they describe the vector field

$$\hat{e}_2 = a(x,y)\frac{\partial}{\partial x} + b(x,y)\frac{\partial}{\partial y}.$$

In other words, there exists the α satisfying the *Codazzi-like* equation

$$\alpha_{xx} + 6\alpha\alpha_x + 4\alpha^3 + c^2\alpha = 0, \tag{1.2}$$

which is a nonlinear ordinary differential equation. In [5], we normalized a and b such that they can be uniquely determined by the function α, and hence we obtained the result that the existence of a constant p-mean curvature surface (without singular points) is equivalent to the existence of a solution to a nonlinear second-order ODE (1.2), which is a kind of *Liénard equations* (cf. [7]). They are one-to-one correspondences in some sense. For a detailed description, see [5, Theorems 1.1, 1.3 and 6.3]. This result tells us that the investigation of the geometry of constant p-mean curvature surfaces in H_1 is equal to the study of the solution of the equation (1.2). More specifically, we obtained a complete set of solutions (see [5, Theorems 1.2 and 1.4] or Theorem 2.1) and used the types of the solutions to the equation to characterize the constant p-mean curvature surfaces as several classes, which are *vertical*, *special type I*, *special type II* and *general type* (see [5, Definitions 5.1 and 5.2] for p-minimal cases and see Definitions 2.2 and 2.3 for the cases with $c \neq 0$ in the present article). After the process of normalization, we obtained a complete set of invariants from the normal form of the α-function. It is worth of our mention that these invariants in some sense measure how different a constant p-mean curvature surface is from the model case, which is the horizontal plane in the p-minimal case, and the Pansu sphere in the case $c > 0$.

We first study rotationally invariant surfaces in H_1 with constant p-mean curvature $H = c$ using the Codazzi-like equation (1.2). In [8], M. Ritoré and C. Rosales made an investigation on such kinds of surfaces by a first-order ODE system. In the present paper, we shall study them again from the point of view of our theory established in the previous paper [5] and the present one. Let $\Sigma(s,\theta)$ be a rotationally invariant surface in H_1 with $H = c$, generated by the curve $\gamma(s) = (x(s), 0, t(s))$, $x(s) \geq 0$, on the xt-plane, that is, Σ is parametrized by

$$\Sigma(s,\theta) = (x(s)\cos\theta, x(s)\sin\theta, t(s)),$$

where $x'^2 + t'^2 = 1$. Here $'$ means taking a derivative with respect to s. Recall the energy

$$E = \frac{xt'}{\sqrt{x^2x'^2 + t'^2}} + \lambda x^2, \tag{1.3}$$

which was introduced in [8] and was shown to be a constant. Here $2\lambda = c$ and notice that our p-mean curvature differs from the one defined in [8] by a sign. Hence, we have Theorems A and B as follows.

Theorem A. *A curve $\gamma = (x,t)$ is the generating curve of a rotationally invariant surface Σ in H_1 with $H = c \neq 0$ if and only if $\gamma = (x,t)$ is defined by $x^2 = \frac{k}{c^2} + r\cos(c\tilde{s})$, $t = -\frac{\tilde{s}}{c} - \frac{r}{2}\sin(c\tilde{s})$, up to a constant, for some horizontal arc-length parameter $\tilde{s}$ and some $k, r \in \mathbb{R}$ such that*

$$k \geq 1 \qquad \text{and} \qquad r = \frac{2}{c^2}\sqrt{k-1}.$$

In addition, we have $k = 2cE + 2$. If $r = 0$, then Σ is a cylinder. If $r \neq 0$, then, in terms of normal coordinates $(\bar{s}, \bar{\theta})$, the two invariants for Σ are

$$\zeta_1(\bar{\theta}) = -\frac{2E\bar{\theta}}{cr}, \qquad \text{up to a constant, which is linear on } \bar{\theta},$$
$$\zeta_2(\bar{\theta}) = -\frac{2cE+2}{c^2r}, \qquad \text{which is a constant.}$$

Theorem B. *A curve $\gamma = (x,t)$ is the generating curve of a rotationally invariant p-minimal surface Σ in H_1 if and only if either t is a constant, and hence Σ is a part of the horizontal plane, or $\gamma = (x,t)$ is defined by $x^2 = \tilde{s}^2 + c_2$, $t = m\tilde{s}$, up to a constant, for some horizontal arc-length parameter $\tilde{s}$ and some $c_2, m \in \mathbb{R}$, $m \neq 0$. In addition, we have $E = m$. In terms of normal coordinates $(\bar{s}, \bar{\theta})$, the two invariants for Σ are*

$$\zeta_1(\bar{\theta}) = E\bar{\theta}, \qquad \text{up to a constant, which is linear on } \bar{\theta},$$
$$\zeta_2(\bar{\theta}) = c_2, \qquad \text{which is a constant.}$$

For more interesting examples, in Section 4, we provide an approach to construct a constant p-mean curvature surface. This approach is an analog of the one we performed in the previous paper [5] for p-minimal surfaces. Actually, in [5], we deformed the horizontal plane along a curve $\mathcal{C}(\theta) = (x_1(\theta), x_2(\theta), x_3(\theta))$ to obtain a p-minimal surface. More specifically, in [5, Section 9], depending on a parametrized curve $\mathcal{C}(\theta) = (x_1(\theta), x_2(\theta), x_3(\theta))$ for $\theta \in \mathbb{R}$, we deformed the graph $u = 0$ to obtain a p-minimal surface parametrized by

$$Y(r,\theta) = (x_1(\theta) + r\cos\theta, x_2(\theta) + r\sin\theta, x_3(\theta) + rx_2(\theta)\cos\theta - rx_1(\theta)\sin\theta),$$

for $r \in \mathbb{R}$. It is easy to check that Y is *an immersion* if and only if either $\Theta\big(\mathcal{C}'(\theta)\big) - \big(x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta\big)^2 \neq 0$ or $r + \big(x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta\big) \neq 0$ for all θ. In particular, the surface Y defines a p-minimal surface of *special type I* if the curve $\mathcal{C}$ satisfies

$$x_3'(\theta) + x_1(\theta)x_2'(\theta) - x_2(\theta)x_1'(\theta) - \big(x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta\big)^2 = 0,$$

for all θ. In addition, the corresponding ζ_1-invariant [5, formula (9.9)] reads

$$\zeta_1(\theta) = x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta - \int \big[x_1'(\theta)\cos\theta + x_2'(\theta)\sin\theta\big]\mathrm{d}\theta. \tag{1.4}$$

Similarly, the surface Y defines a p-minimal surface of *general type* if the curve $\mathcal{C}$ satisfies

$$x_3'(\theta) + x_1(\theta)x_2'(\theta) - x_2(\theta)x_1'(\theta) - \big(x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta\big)^2 \neq 0,$$

for all θ. In addition, the corresponding ζ_1- and ζ_2-invariant read

$$\begin{aligned}\zeta_1(\theta) &= x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta - \int \big[x_1'(\theta)\cos\theta + x_2'(\theta)\sin\theta\big]\mathrm{d}\theta,\\ \zeta_2(\theta) &= x_3'(\theta) + x_1(\theta)x_2'(\theta) - x_2(\theta)x_1'(\theta) - \big(x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta\big)^2.\end{aligned} \tag{1.5}$$

In Section 4, we construct a constant p-mean curvature surface by perturbing the Pansu sphere along a given curve $\mathcal{C}(\theta)$. In Section 2.1, we see that the Pansu sphere (2.2) can be parametrized by

$$X(s,\theta) = \begin{pmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}x(s)\\ y(s)\\ t(s)\end{pmatrix} = \begin{pmatrix}x(s)\cos\theta - y(s)\sin\theta\\ x(s)\sin\theta + y(s)\cos\theta\\ t(s)\end{pmatrix},$$

where

$$\begin{aligned}x(s) = \frac{1}{2\lambda}\sin(2\lambda s), \qquad y(s) = -\frac{1}{2\lambda}\cos(2\lambda s) + \frac{1}{2\lambda},\\ t(s) = \frac{1}{4\lambda^2}\sin(2\lambda s) - \frac{1}{2\lambda}s + \frac{\pi}{4\lambda^2},\end{aligned}$$

with $X(0,\theta) = \big(0, 0, \frac{\pi}{4\lambda^2}\big)$ and $X\big(\frac{\pi}{\lambda}, \theta\big) = \big(0, 0, -\frac{\pi}{4\lambda^2}\big)$ as the North pole and South pole, respectively. We deform it along $\mathcal{C}(\theta) = (x_1(\theta), x_2(\theta), x_3(\theta))$ to obtain a constant p-mean curvature surface $Y(s,\theta)$ as follows

$$\begin{aligned}Y(s,\theta) = (x_1(\theta) + (x(s)\cos\theta - y(s)\sin\theta), x_2(\theta) + (x(s)\sin\theta + y(s)\cos\theta),\\ x_3(\theta) + t(s) + x_2(\theta)(x(s)\cos\theta - y(s)\sin\theta) - x_1(\theta)(x(s)\sin\theta + y(s)\cos\theta)).\end{aligned}$$

We also give a condition for Y to be an immersion. The coordinate system (s,θ) for Y is a compatible one. We have (see (4.4))

$$\alpha = \lambda\frac{A(\theta)\cos 2\lambda s + \big(\frac{1}{2\lambda} - B(\theta)\big)\sin 2\lambda s}{\big(B(\theta) - \frac{1}{2\lambda}\big)\cos 2\lambda s + A(\theta)\sin 2\lambda s + D(\theta)}, \tag{1.6}$$

where

$$\begin{aligned}A(\theta) = x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta, \qquad B(\theta) = x_2'(\theta)\sin\theta + x_1'(\theta)\cos\theta,\\ D(\theta) = \lambda\Theta(\mathcal{C}'(\theta)) + \left(\frac{1}{2\lambda} - B(\theta)\right), \qquad V(\theta) = \left(A(\theta), \frac{1}{2\lambda} - B(\theta)\right).\end{aligned}$$

It is obvious that $V(\theta) = 0$ implies $\alpha = 0$, and hence Y is a cylinder. For nonzero $V(\theta)$, we then define

$$\|V(\theta)\| = \sqrt{[A(\theta)]^2 + \left[\frac{1}{2\lambda} - B(\theta)\right]^2}$$

and write

$$G(\theta) = \frac{D(\theta)}{\|V(\theta)\|}, \qquad \frac{V(\theta)}{\|V(\theta)\|} = (\sin\zeta(\theta), \cos\zeta(\theta)),$$

for some function $\zeta(\theta)$. From (1.6), we thus have

$$\alpha = \lambda\frac{\sin\big(2\lambda s + \zeta(\theta)\big)}{G(\theta) - \cos\big(2\lambda s + \zeta(\theta)\big)}.$$

Finally, we normalize the α-function to obtain the two invariants for Y, stated in Theorem 4.1. We list it here as the third main result of the present paper.

Theorem C. *If $V = 0$, then $\alpha = 0$. If $V \neq 0$, we consider the new coordinates $\tilde{s} = s + \Gamma(\theta)$, $\tilde{\theta} = \Psi(\theta)$, where*

$$\Gamma(\theta) = \frac{\theta}{2\lambda} - \int D(\theta)\mathrm{d}\theta, \qquad \Psi(\theta) = 2\lambda \int \|V(\theta)\|\,\mathrm{d}\theta,$$

then the coordinate system $\big(\tilde{s}, \tilde{\theta}\big)$ is normal. In terms of the normal coordinates, the invariants of Y are given by

$$\zeta_1\big(\tilde{\theta}\big) = \zeta(\theta) - 2\lambda\Gamma(\theta), \qquad \zeta_2\big(\tilde{\theta}\big) = G(\theta), \tag{1.7}$$

where $\theta = \Psi^{-1}\big(\tilde{\theta}\big)$.

In Section 5, we use formula (1.4), (1.5) and (1.7) to construct various examples of surfaces of constant p-mean curvature including degenerate p-minimal surfaces of special type I.

2. Solutions to the Codazzi-like equation

The Codazzi-like equation for a surface in H_1 with constant p-mean curvature $H = c > 0$ is

$$\alpha_{xx} + 6\alpha\alpha_x + 4\alpha^3 + c^2\alpha = 0. \tag{2.1}$$

Theorem 2.1 ([5])**.** *Besides the following three special solutions to* (2.1)*,*

$$\alpha(x) = 0, \qquad -\frac{c}{2}\tan(cx + cK_1), \qquad \alpha(x) = -\frac{c}{2}\tan\Big(\frac{c}{2}x - \frac{c}{2}K_2\Big),$$

we have the general solution to (2.1) *of the form*

$$\alpha(x) = \frac{c}{2}\frac{\sin\big(cx + c_1\big)}{c_2 - \cos\big(cx + c_1\big)},$$

which depends on constants K_1, K_2, c_1, and c_2.

Note that all the solutions are a periodic function with $\alpha\big(x + \frac{2\pi}{c}\big) = \alpha(x)$ for all x. We give some remarks as follows.

(1) In terms of the following identities

$$-\tan\Big(\theta + \frac{\pi}{2}\Big) = \cot\theta = \frac{\sin 2\theta}{1 - \cos 2\theta}, \qquad -\tan 2\theta = -\frac{\sin 2\theta}{\cos 2\theta} = \frac{\sin 2\theta}{0 - \cos 2\theta},$$

we see that the two nontrivial special solutions in Theorem 2.1 correspond to the general solution in Theorem 2.1 with $c_2 = 0$ and $c_2 = 1$, respectively.

(2) From the following identity

$$\frac{\sin(\theta + \pi)}{c_2 - \cos(\theta + \pi)} = \frac{\sin\theta}{-c_2 - \cos\theta},$$

we can assume without loss of generality that $c_2 \geq 0$ in the general solution.

Due to Theorem 2.1, we are able to use the types of the solutions to (2.1) to classify the constant p-mean curvature surfaces into several classes, which are vertical, special type I, special type II and general type. In terms of compatible coordinates (x, y), the function $\alpha(x, y)$ is a solution to the *Codazzi-like* equation (2.1) for any given y. By Theorem 2.1, the function $\alpha(x, y)$ hence has one of the following forms of special types

$$0, \qquad \frac{c}{2}\frac{\sin(cx + c_1)}{0 - \cos(cx + c_1)}, \qquad \frac{c}{2}\frac{\sin\big(cx + c_1\big)}{1 - \cos\big(cx + c_1\big)},$$

and general types $\frac{c}{2}\frac{\sin(cx+c_1)}{c_2-\cos(c(x+c_1)}$, where, instead of constants, both c_1 and c_2 are now functions of y. Notice that it is convenient at some point to assume that $c_2(y) \geq 0$ for all y. We now use the types of the function $\alpha(x, y)$ to define the types of constant p-mean curvature surfaces as follows.

Definition 2.2. Locally, we say that a constant p-mean curvature surface is

(1) *vertical* if α vanishes (i.e., $\alpha(x, y) = 0$ for all x, y);
(2) of *special type I* if $\alpha = \frac{c}{2}\frac{\sin(cx+c_1(y))}{1-\cos(cx+c_1(y))}$;
(3) of *special type II* if $\alpha = \frac{c}{2}\frac{\sin(cx+c_1(y))}{0-\cos(cx+c_1(y))}$;
(4) of *general type* if $\alpha = \frac{c}{2}\frac{\sin(cx+c_1(y))}{c_2(y)-\cos(cx+c_1(y))}$ with $c_2(y) \notin \{0, 1\}$ for all y.

We further divide constant p-mean curvature surfaces of general type into three classes as follows.

Definition 2.3. A constant p-mean curvature surface of *general type* is

(1) of *type I* if $c_2(y) > 1$ for all y;

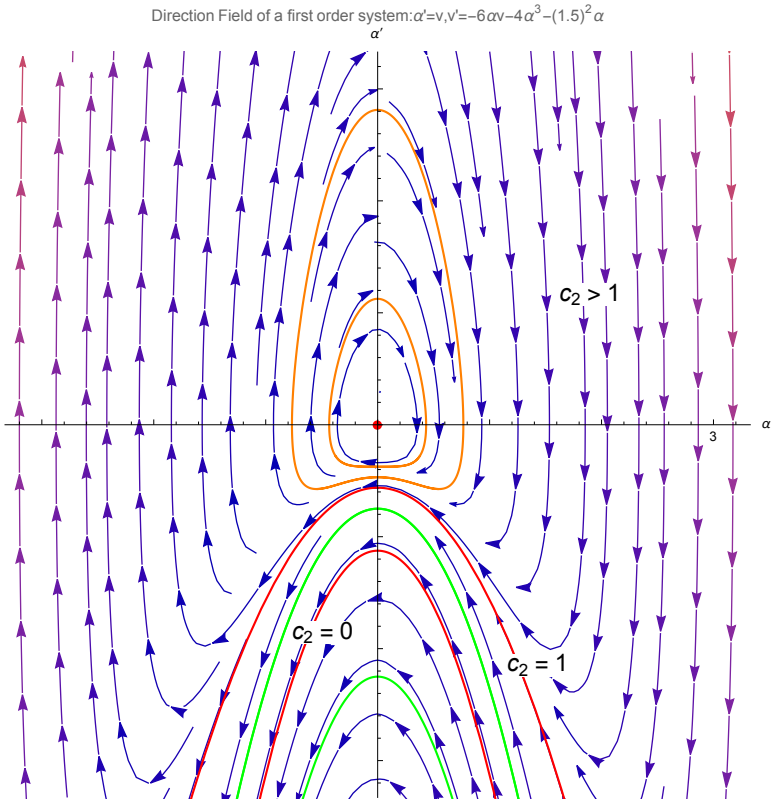


FIGURE 1. Direction field for $c = 1.5$.

(2) of *type II* if $0 < c_2(y) < 1$ for all y, and $\frac{-c_1+\cos^{-1} c_2}{c} < x < \frac{2\pi-c_1-\cos^{-1} c_2}{c}$;
(3) of *type III* if $0 < c_2(y) < 1$ for all y, and either $\frac{-c_1}{c} \leq x < \frac{-c_1+\cos^{-1} c_2}{c}$ or $\frac{2\pi-c_1-\cos^{-1} c_2}{c} < x \leq \frac{-c_1+2\pi}{c}$,

where $\cos^{-1}$ is the inverse of the function $\cos\colon [0,\pi] \to [-1,1]$.

We notice that the *type* is invariant under the action of a Heisenberg rigid motion and the regular part of a constant p-mean curvature surface $\Sigma \subset H_1$ is a union of these types of surfaces. The corresponding paths of each type of α are shown on the phase plane (see Figure 1). We express some basic facts as follows.

- If α vanishes, then it is part of a vertical cylinder.
- The two concave downward parabolas in red represent

$$\alpha = \frac{c}{2}\frac{\sin(cx+c_1)}{1-\cos(cx+c_1)}, \qquad \frac{c}{2}\frac{\sin(cx+c_1)}{0-\cos(cx+c_1)},$$

respectively. The one for $\alpha = \frac{c}{2}\frac{\sin(cx+c_1)}{1-\cos(cx+c_1)}$ is above the one for $\alpha = \frac{c}{2}\frac{\sin(cx+c_1)}{0-\cos(cx+c_1)}$. For surfaces of *special type I*, we have that $\alpha\left(\frac{\pi-c_1}{c}\right) = 0$, $\alpha'\left(\frac{\pi-c_1}{c}\right) = -\frac{c^2}{4}$ and

$$\alpha \to \begin{cases} \infty, & \text{if } x \to \dfrac{-c_1}{c} \quad \text{from the right,} \\ -\infty, & \text{if } x \to \dfrac{2\pi-c_1}{c} \quad \text{from the left,} \end{cases}$$

and, for surfaces of *special type II* in which α has period π, we have that

$$\alpha\left(\frac{-c_1}{c}\right) = \alpha\left(\frac{\pi-c_1}{c}\right) = 0, \qquad \alpha'\left(\frac{-c_1}{c}\right) = \alpha'\left(\frac{\pi-c_1}{c}\right) = -\frac{c^2}{2}$$

and

$$\alpha \to \begin{cases} \infty, & \text{if } x \to \dfrac{\pi-2c_1}{2c} \quad \text{from the right,} \\ -\infty, & \text{if } x \to \dfrac{3\pi-2c_1}{2c} \quad \text{from the left.} \end{cases}$$

- The closed curves in orange on the phase plane correspond to the family of solutions

$$\alpha(x) = \frac{c}{2}\frac{\sin(cx+c_1)}{c_2-\cos(cx+c_1)},$$

where c_1, c_2 are constants and $c_2 > 1$, which are of *type I*. There exist zeros for α-function at $x = \frac{-c_1}{c}$, $\frac{\pi - c_1}{c}$, at which we have that

$$\alpha'\left(\frac{-c_1}{c}\right) = \frac{1}{2}\frac{c^2}{c_2 - 1} > 0, \qquad \alpha'\left(\frac{\pi - c_1}{c}\right) = -\frac{1}{2}\frac{c^2}{c_2 + 1} < 0.$$

There are no singular points for surfaces of *type I*.

- The curves in between the two red concave downward parabolas are of *type II*. The α-function of *type II* has a zero at $x = \frac{\pi - c_1}{c}$, and

$$\alpha'\left(\frac{\pi - c_1}{c}\right) = -\frac{1}{2}\frac{c^2}{c_2 + 1} < 0.$$

For surfaces of *type II*, it can be checked that

$$\alpha \to \begin{cases} \infty, & \text{if } x \to \dfrac{-c_1 + \cos^{-1} c_2}{c} \quad \text{from the right,} \\ -\infty, & \text{if } x \to \dfrac{-c_1 + 2\pi - \cos^{-1} c_2}{c} \quad \text{from the left.} \end{cases}$$

- The curves beneath the lower concave downward parabola are of *type III*. There exists a zero for α-function at $x = \frac{-c_1}{c}$, and

$$\alpha'\left(\frac{-c_1}{c}\right) = \frac{1}{2}\frac{c^2}{c_2 - 1} < 0.$$

For surfaces of *type III*, we have

$$\alpha \to \begin{cases} -\infty, & \text{if } x \to \dfrac{-c_1 + \cos^{-1} c_2}{c} \quad \text{from the left,} \\ \infty, & \text{if } x \to \dfrac{-c_1 + 2\pi - \cos^{-1} c_2}{c} \quad \text{from the right.} \end{cases}$$

2.1. **The Pansu sphere.**

Lemma 2.4 (Pansu sphere)**.** *A Pansu sphere given in* [1] *by*

$$f(z) = \frac{1}{2\lambda^2}\big(\lambda|z|\sqrt{1 - \lambda^2|z|^2} + \cos^{-1}(\lambda|z|)\big), \qquad |z| \le \frac{1}{\lambda}, \tag{2.2}$$

of constant p-mean curvature $c = 2\lambda$ has its α-function of special type I. In fact, we have

$$\alpha = \frac{\lambda \sin(2\lambda s)}{(1 - \cos(2\lambda s))}, \qquad a = \frac{-\lambda}{\sqrt{1 + \alpha^2}}, \qquad b = \frac{2\lambda^2}{\sqrt{1 + \alpha^2}(1 - \cos 2\lambda s)}.$$

Proof. We parametrize a Pansu sphere by

$$X(s, \theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x(s) \\ y(s) \\ t(s) \end{pmatrix} = \begin{pmatrix} x(s)\cos\theta - y(s)\sin\theta \\ x(s)\sin\theta + y(s)\cos\theta \\ t(s) \end{pmatrix}, \tag{2.3}$$

where

$$\begin{aligned} x(s) &= \frac{1}{2\lambda}\sin(2\lambda s), \qquad y(s) = -\frac{1}{2\lambda}\cos(2\lambda s) + \frac{1}{2\lambda}, \\ t(s) &= \frac{1}{4\lambda^2}\sin(2\lambda s) - \frac{1}{2\lambda}s + \frac{\pi}{4\lambda^2}, \end{aligned} \tag{2.4}$$

with $X(0, \theta) = \big(0, 0, \frac{\pi}{4\lambda^2}\big)$ and $X\big(\frac{\pi}{\lambda}, \theta\big) = \big(0, 0, -\frac{\pi}{4\lambda^2}\big)$ as the North pole and South pole, respectively. We then have $e_1 := X_s$, which means that $X(s, \theta)$ defines a compatible coordinate system. Moreover, $e_2 = Je_1$, where

$$\begin{aligned} e_1 = X_s &= \big(x'(s)\cos\theta - y'(s)\sin\theta, x'(s)\sin\theta + y'(s)\cos\theta, t'(s)\big) \\ &= \big(x'(s)\cos\theta - y'(s)\sin\theta\big)\mathring{e}_1 + \big(x'(s)\sin\theta + y'(s)\cos\theta\big)\mathring{e}_2, \\ e_2 &= -\big(x'(s)\sin\theta + y'(s)\cos\theta\big)\mathring{e}_1 + \big(x'(s)\cos\theta - y'(s)\sin\theta\big)\mathring{e}_2. \end{aligned}$$

We note that α is a function satisfying

$$\alpha e_2 + T = \mathcal{A}X_s + \mathcal{B}X_\theta, \tag{2.5}$$

for some functions $\mathcal{A}$ and $\mathcal{B}$. Direct calculation shows that

$$X_\theta = (-x(s)\sin\theta - y(s)\cos\theta)\mathring{e}_1 + (x(s)\cos\theta - y(s)\sin\theta)\mathring{e}_2 + (x^2(s) + y^2(s))T.$$

Therefore, (2.5) implies

$$\begin{aligned}
-\alpha\big(x'(s)\sin\theta + y'(s)\cos\theta\big) &= \mathcal{A}\big(x'(s)\cos\theta - y'(s)\sin\theta\big) + \mathcal{B}(-x(s)\sin\theta - y(s)\cos\theta),\\
\alpha\big(x'(s)\cos\theta - y'(s)\sin\theta\big) &= \mathcal{A}\big(x'(s)\sin\theta + y'(s)\cos\theta\big) + \mathcal{B}(x(s)\cos\theta - y(s)\sin\theta),\\
1 &= \mathcal{B}\big(x^2(s) + y^2(s)\big). \qquad (2.6)
\end{aligned}$$

The last equation of (2.6) yields $\mathcal{B} = \frac{1}{x^2(s)+y^2(s)}$, and hence

$$b = \frac{\mathcal{B}}{\sqrt{1+\alpha^2}} = \frac{2\lambda^2}{\sqrt{1+\alpha^2}(1-\cos 2\lambda s)}.$$

The first two equations of (2.6) indicate $-\alpha\big(\big(x'^2\big) + \big(y'^2\big)\big) = \mathcal{B}\big(-xx' - yy'\big)$, which implies

$$\alpha = \frac{xx' + yy'}{x^2 + y^2}. \qquad (2.7)$$

In what follows, we claim the above α is one of special solutions. Notice that (2.4) shows

$$x^2 + y^2 = \frac{1}{4\lambda^2}(2 - 2\cos(2\lambda s)),$$

and hence (2.7) can be rewritten as

$$\alpha = \frac{1}{2}\big(\ln x^2 + y^2\big)' = \frac{\lambda \sin(2\lambda s)}{(1-\cos(2\lambda s))}.$$

Substituting b and α into (2.6), we have $a = \frac{\mathcal{A}}{\sqrt{1+\alpha^2}} = \frac{-\lambda}{\sqrt{1+\alpha^2}}$. □

Given a α-function, we have shown [5] that the first fundamental form (a, b) is determined up to two functions $h(y)$ and $k(y)$ as follows.

Proposition 2.5. *For any $\alpha(x,y) = \frac{c}{2}\frac{\sin(cx+c_1(y))}{c_2(y)-\cos(cx+c_1(y))}$, the explicit formula for the induced metric on a constant p-mean curvature surface with c as its p-mean curvature and this α as its α-function is given by*

$$a = \left(-\frac{c}{2} + \frac{\frac{c}{2}c_2}{(c_2 - \cos(cx+c_1))} + \frac{h(y)}{|c_2 - \cos(cx+c_1)|}\right)\frac{1}{\big(1+\alpha^2\big)^{1/2}},$$

and

$$b = \left(\frac{e^{k(y)}}{|c_2 - \cos(cx+c_1)|}\right)\frac{1}{\big(1+\alpha^2\big)^{1/2}},$$

for some functions $h(y)$ and $k(y)$.

2.2. **The normalization.** As we normalize the induced metric a and b to be close as much as possible to the metric induced on the horizontal p-minimal plane, we would like to normalize a and b so that they look like the induced metric of the Pansu sphere. Indeed, from the transformation law [5, formula (2.20)], it is easy to see that there exist another compatible coordinates $(\tilde{x}, \tilde{y})$, called normal coordinates such that

$$\tilde{a} = -\frac{\frac{c}{2}}{\big(1+\alpha^2\big)^{1/2}}, \qquad (2.8)$$

$$\tilde{b} = \frac{\frac{c^2}{2}}{|c_2 - \cos(cx+c_1)|\big(1+\alpha^2\big)^{1/2}}, \qquad (2.9)$$

where $c = 2\lambda$. Such normal coordinates are uniquely determined up to a translation. We thus have the following theorem.

Theorem 2.6. *In normal coordinates (x, y), the functions $c_1(y)$ and $c_2(y)$ in the expression $\alpha(x) = \frac{c}{2}\frac{\sin(cx+c_1)}{c_2-\cos(cx+c_1)}$ are unique in the following sense: up to a translation on y, $c_2(y)$ is unique, and $c_1(y)$ is unique up to a constant. We denote these two unique functions by $\zeta_1(y) = c_1(y)$, $\zeta_2(y) = c_2(y)$. Therefore, the set $\{\zeta_1(y), \zeta_2(y)\}$ constitutes a* complete *set of invariants for those surfaces (α not vanishing).*

It is worth our attention that, for the surfaces with $c_2 > 1$, the denominator of the formula for α is never zero. That means the surfaces won't extend to a surface with singular points. Moreover, if the surface is closed, it must be a closed constant p-mean curvature surface without singular points, which means the surface is of type of torus. This indicates that it is possible to find a Wente-type torus in this class of surfaces.

2.3. **The structure of the singular sets.** In this subsection, we study the structure of the singular set. For the general type, we choose a normal coordinate system (s,θ) such that

$$\alpha=\lambda\frac{\sin\left(2\lambda s+\zeta_1(\theta)\right)}{\zeta_2(\theta)-\cos\left(2\lambda s+\zeta_1(\theta)\right)},$$

and

$$a=-\frac{\lambda}{\sqrt{1+\alpha^2}},\qquad b=\frac{2\lambda^2}{\left|\zeta_2(\theta)-\cos\left(2\lambda s+\zeta_1(\theta)\right)\right|\sqrt{1+\alpha^2}}.$$

Then the singular set is the graph of the function $x(\theta)=\frac{\cos^{-1}(\zeta_2(\theta))-\zeta_1(\theta)}{2\lambda}$. The induced metric I (or the first fundamental form) on the regular part reads

$$I=\mathrm{d}s\otimes\mathrm{d}s-\frac{a}{b}\mathrm{d}s\otimes\mathrm{d}\theta-\frac{a}{b}\mathrm{d}\theta\otimes\mathrm{d}s+\frac{\left(1+a^2\right)}{b^2}\mathrm{d}\theta\otimes\mathrm{d}\theta.$$

Now we use the metric to compute the length of the singular set $\left\{\left(\frac{\cos^{-1}(\zeta_2(\theta))-\zeta_1(\theta)}{2\lambda},\theta\right)\right\}$, where θ belongs to some open interval.

Case $\zeta_2(\theta)\neq 1$. Let $\gamma(\theta)=\left(\frac{\cos^{-1}(\zeta_2(\theta))-\zeta_1(\theta)}{2\lambda},\theta\right)$, which is a parametrization of the singular set. Then the square of the velocity at θ is

$$|\gamma'(\theta)|^2=[q'(\theta)]^2-\frac{2aq'(\theta)}{b}+\frac{a^2+1}{b^2}=\frac{[a-bq'(\theta)]^2+1}{b^2}>0\qquad\text{for all}\quad\theta,\tag{2.10}$$

where

$$q'(\theta)=\left[\frac{\cos^{-1}(\zeta_2(\theta))-\zeta_1(\theta)}{2\lambda}\right]'=\frac{-\zeta_2'(\theta)}{2\lambda\sqrt{1-\zeta_2^2(\theta)}}-\frac{\zeta_1'(\theta)}{2\lambda}.$$

Formula (2.10) shows that the parametrized curve $\gamma(\theta)$ of the singular set has a positive length.

Case $\zeta_2(\theta)=1$. We parametrize the singular set by $\gamma(\theta)=\left(\frac{\cos^{-1}(1-\epsilon)-\zeta_1(\theta)}{2\lambda},\theta\right)$ for $\epsilon>0$. It is easy to see $\gamma'(\theta)=\left(\frac{-\zeta_1'(\theta)}{2\lambda},1\right)$. When $\varepsilon\to 0$, the metric

$$I=\mathrm{d}s\otimes\mathrm{d}s-\frac{a}{b}\mathrm{d}s\otimes\mathrm{d}\theta-\frac{a}{b}\mathrm{d}\theta\otimes\mathrm{d}s+\frac{\left(1+a^2\right)}{b^2}\mathrm{d}\theta\otimes\mathrm{d}\theta$$

degenerates to $\tilde{I}=\mathrm{d}s\otimes\mathrm{d}s$.

Then the square of the velocity at θ is $\left|\gamma'(\theta)\right|^2=\frac{[\zeta_1'(\theta)]^2}{4\lambda^2}$. Thus, if $\zeta_1(\theta)=c_1$ and $\zeta_2(\theta)=1$, the length of the parametrized curve $\gamma(\theta)$ of the singular set is zero. This result coincides with the singular set for the Pansu sphere being isolated. We conclude the above discussion with the following theorem, an analog of [5, Theorem 1.7].

Theorem 2.7. *The singular set of a constant p-mean surface with $H=c\neq 0$ is either*

(1) *an isolated point; or*
(2) *a smooth curve.*

In addition, an isolated singular point only happens on the surfaces of special type I with $\zeta_1=$ const, namely, a part of the Pansu sphere containing one of the poles as the isolated singular point.

Theorem 2.7 together with [5, Theorem 1.7] are just special cases of [2, Theorem 3.3]. However, we give a computable proof of this result for constant p-mean surfaces. We also have the description of how a characteristic leaf goes through a singular curve, which is called a "go through" theorem in [2]. Suppose p_0 is a point in a singular curve. From the above basic facts, we see that a characteristic curve γ always reaches the singular point p_0 going a finite distance. From the opposite direction, suppose $\tilde{\gamma}$ is another characteristic curve that reaches p_0. Then the union of γ, p_0 and $\tilde{\gamma}$ forms a smooth curve (we also refer the reader to the proof of [5, Theorem 1.8], they are similar). We thus have the following theorem.

Theorem 2.8. *Let $\Sigma\subset H_1$ be a constant p-mean surface with $H=c\neq 0$. Then the characteristic foliation is smooth around the singular curve in the following sense that each leaf can be extended smoothly to a point on the singular curve.*

Making use of Theorem 2.8, we have the following result.

Theorem 2.9. *Let Σ be a constant p-mean surface of type II (III) with $H=c\neq 0$. If it can be smoothly extended through the singular curve, then the other side of the singular curve is of type III (II).*

Therefore, we see that a surface of general type II is always pasted together with a surface of general type III at a singular curve and vice versa.

3. Rotationally invariant surfaces in H_1

Let $\Sigma(s,\theta)$ be a rotationally invariant surface in H_1 generated by a curve $\gamma(s)=(x(s),0,t(s))$ on the xt-plane, that is, Σ is parametrized by $\Sigma(s,\theta)=(x(s)\cos\theta,x(s)\sin\theta,t(s))$, where $x'^2+t'^2=1$. Here $'$ means taking a derivative with respect to s.

3.1. **The computation of $\boldsymbol{H}$, $\boldsymbol{\alpha}$, $\boldsymbol{a}$ and $\boldsymbol{b}$.** Now we consider the horizontal (see [3, Definition 1.1]) generating curve

$$\tilde\gamma(s) = (x(s)\cos\theta(s), x(s)\sin\theta(s), t(s)).$$

Lemma 3.1. *$\tilde\gamma$ is horizontal if and only if $t' + x^2\theta' = 0$.*

Proof. Note that at the point $\tilde\gamma(s)$,

$$\mathring{e}_1 = \frac{\partial}{\partial x_1} + y_1\frac{\partial}{\partial z} = \frac{\partial}{\partial x_1} + x(s)\sin\theta(s)\frac{\partial}{\partial z},$$
$$\mathring{e}_2 = \frac{\partial}{\partial y_1} - x_1\frac{\partial}{\partial z} = \frac{\partial}{\partial y_1} - x(s)\cos\theta(s)\frac{\partial}{\partial z},$$

and direct computations imply

$$\tilde\gamma'(s) = \big(x'\cos\theta - x\theta'\sin\theta\big)\mathring{e}_1 + \big(x'\sin\theta + x\theta'\cos\theta\big)\mathring{e}_2 + \big(t' + x^2\theta'\big)T,$$

and hence $\tilde\gamma'(s) \in \xi$ if and only if $t' + x^2\theta' = 0$. □

Let $\tilde s$ be the horizontal arc-length of $\tilde\gamma(s)$. We can thus re-parametrize the surface $\Sigma(s,\theta)$ to be

$$\Sigma(\tilde s, \tilde\theta) = \big(x(s)\cos\theta(s)\cos\tilde\theta - x(s)\sin\theta(s)\sin\tilde\theta, x(s)\cos\theta(s)\sin\tilde\theta + x(s)\sin\theta(s)\cos\tilde\theta, t(s)\big),$$

with a compatible coordinate system

$$e_1 = \frac{\partial}{\partial\tilde s} = \Sigma_s\frac{\partial s}{\partial\tilde s}, \qquad \text{where} \quad \Sigma_s = \big(x'\cos\phi - x\theta'\sin\phi\big)\mathring{e}_1 + \big(x'\sin\phi + x\theta'\cos\phi\big)\mathring{e}_2.$$

Moreover, we see $|\tilde\gamma|'^2 = \frac{x^2x'^2+t'^2}{x^2}$, so that we may choose $\tilde s$ such that $\big|\frac{\mathrm{d}\tilde\gamma(\tilde s)}{d\tilde s}\big| = 1$, that is,

$$\frac{\mathrm{d}\tilde s}{\mathrm{d}s} = |\tilde\gamma'(s)| = \frac{\sqrt{x^2x'^2 + t'^2}}{x}. \tag{3.1}$$

Manipulating Σ to be

$$\Sigma\big(\tilde s, \tilde\theta\big) = \big(x(s)\cos\big(\theta(s)+\tilde\theta\big), x(s)\sin\big(\theta(s)+\tilde\theta\big), t(s)\big), \qquad \big(\text{denote } \phi = \theta(s) + \tilde\theta\big)$$

and obtain

$$\Sigma_{\tilde s} = \frac{\mathrm{d}s}{\mathrm{d}\tilde s}\big(x'\cos\phi - x\theta'\sin\phi\big)\mathring{e}_1 + \frac{\mathrm{d}s}{\mathrm{d}\tilde s}\big(x'\sin\phi + x\theta'\cos\phi\big)\mathring{e}_2,$$
$$\Sigma_{\tilde\theta} = -x\sin\phi\mathring{e}_1 + x\cos\phi\mathring{e}_2 + x^2\frac{\partial}{\partial z}.$$

Then

$$e_1 = \Sigma_{\tilde s} = \frac{x}{\sqrt{x^2x'^2+t'^2}}\big(x'\cos\phi - x\theta'\sin\phi\big)\mathring{e}_1 + \frac{\mathrm{d}s}{\mathrm{d}\tilde s}\big(x'\sin\phi + x\theta'\cos\phi\big)\mathring{e}_2, \tag{3.2}$$

$$e_2 = Je_1 = \frac{x}{\sqrt{x^2x'^2+t'^2}}\big(x'\cos\phi - x\theta'\sin\phi\big)\mathring{e}_2 - \frac{\mathrm{d}s}{\mathrm{d}\tilde s}\big(x'\sin\phi + x\theta'\cos\phi\big)\mathring{e}_1. \tag{3.3}$$

The fact that $\frac{\alpha e_2 + T}{\sqrt{1+\alpha^2}} \in T\Sigma$ implies $\alpha e_2 + T = a\sqrt{1+\alpha^2}\Sigma_{\tilde s} + b\sqrt{1+\alpha^2}\Sigma_{\tilde\theta}$. Using (3.2), (3.3) and comparing the coefficients of $\mathring{e}_1$, $\mathring{e}_2$, and T, respectively, one sees

$$a = \frac{t'}{x\sqrt{1+\alpha^2}\sqrt{x^2x'^2+t'^2}}, \qquad b = \frac{1}{x^2\sqrt{1+\alpha^2}}, \qquad \alpha = \frac{x'}{\sqrt{x^2x'^2+t'^2}}, \tag{3.4}$$

and hence, from the first equation of the integrability conditions (1.1), we have

$$H = -\frac{x^3\big(x't'' - x''t'\big) + t'^3}{x\big\{x^2x'^2 + t'^2\big\}^{3/2}}. \tag{3.5}$$

3.2. **Another understanding of energy $\boldsymbol{E}$.** In this subsection, we assume moreover that the rotationally invariant surface Σ is of constant p-mean curvature. We consider the relation between the integrability condition and the energy discussed in Ritoré and Rosales' paper [8]. The integrability condition $a_{\tilde s} - \frac{a}{b}b_{\tilde s} + \frac{c\alpha}{\sqrt{1+\alpha^2}} = 0$ indicates that

$$\int\left(\frac{a_{\tilde s}}{b} - \frac{a}{b^2}b_{\tilde s} + \frac{c\alpha}{b\sqrt{1+\alpha^2}}\right)\mathrm{d}\tilde s \tag{3.6}$$

is a constant. Then we have (3.6) computed as

$$\frac{a}{b} + c\int\frac{x^2x'}{\sqrt{x^2x'^2+t'^2}}\mathrm{d}\tilde s = \frac{a}{b} + c\int xx'\mathrm{d}s = \frac{a}{b} + \lambda x^2, \qquad \text{up to a constant,}$$

which clearly says that $\frac{a}{b}+\lambda x^2$ is constant. The constant $\frac{a}{b}+\lambda x^2$ interprets the energy E based on Ritoré's discussion. Indeed, we have

$$E=\frac{a}{b}+\lambda x^2=\frac{xt'}{\sqrt{x^2x'^2+t'^2}}+\lambda x^2=t_{\tilde{s}}+\lambda x^2, \tag{3.7}$$

that is, $t_{\tilde{s}}=E-\lambda x^2$. One sees

$$t=E\tilde{s}-\lambda\int x^2\mathrm{d}\tilde{s}. \tag{3.8}$$

3.3. **The Coddazi-like equation.** For later use, we calculate $1+\alpha^2=\frac{1+x^2x'^2}{x^2x'^2+t'^2}$ and convert α to be of the general form

$$\alpha=\frac{x'}{\sqrt{x^2x'^2+t'^2}}=\frac{x_{\tilde{s}}}{x}. \tag{3.9}$$

Note that α satisfies the Coddazi-like equation $\alpha_{\tilde{s}\tilde{s}}+6\alpha\alpha_{\tilde{s}}+4\alpha^3+c^2\alpha=0$, where $c=2\lambda$. Then this ODE immediately shows

$$\frac{x_{\tilde{s}\tilde{s}\tilde{s}}}{x}+3\frac{x_{\tilde{s}}x_{\tilde{s}\tilde{s}}}{x^2}+c^2\frac{x_{\tilde{s}}}{x}=0. \tag{3.10}$$

The equation (3.10) is manipulated to be $\big(xx_{\tilde{s}\tilde{s}}+(x_{\tilde{s}})^2+\frac{c^2}{2}x^2\big)_{\tilde{s}}=0$, which gives

$$\big(x^2\big)_{\tilde{s}\tilde{s}}+c^2x^2=k,\qquad \text{for some constant } k. \tag{3.11}$$

Let $u=x^2$, then (3.11) becomes a second-order inhomogeneous constant coefficient ODE

$$u_{\tilde{s}\tilde{s}}+c^2u=k. \tag{3.12}$$

(I) Suppose $c\neq 0$, the homogeneous ODE $u_{\tilde{s}\tilde{s}}+c^2u=0$ has the general solution u_h given by

$$u_h=k_1\sin(c\tilde{s})+k_2\cos(c\tilde{s})=r\cos(c\tilde{s}-c_1), \tag{3.13}$$

where $r=\sqrt{k_1^2+k_2^2}$ and $r\sin c_1=k_1$. One also notes that $u_p=\frac{k}{c^2}$ is a particular solution to (3.12), and hence

$$x^2=u=\frac{k}{c^2}+r\cos(c\tilde{s}-c_1). \tag{3.14}$$

(II) When $c=0$, it is clear that (3.11) becomes $\big(x^2\big)_{\tilde{s}\tilde{s}}=k$, which implies

$$x^2=k\tilde{s}^2+2k_1\tilde{s}+k_2, \tag{3.15}$$

for some constants k, k_1 and k_2.

Example 3.2. If $k=0$, $k_1=0$, then (3.15) yields $x=\sqrt{k_2}$. On the other hand, (3.16) suggests $0=\frac{t'^3}{\sqrt{k_2}t'^3}=\frac{1}{\sqrt{k_2}}>0$, which is a contradiction. We conclude that there are no such kinds of p-minimal surfaces ($k=0$, $k_1=0$) which are rotationally symmetric. In this case, α vanishes so that it corresponds a vertical cylinder surface which is absolutely not p-minimal.

3.4. **The relation between k and E.** Assume that $c=2\lambda\neq 0$. We write (3.5) as

$$\begin{aligned}-2\lambda&=\frac{x^3\big(x't''-x''t'\big)+t'^3}{x\big\{x^2x'^2+t'^2\big\}^{3/2}}\\&=\frac{x^2\big(\frac{t'}{x'}\big)'x'^2}{\big\{x^2x'^2+t'^2\big\}^{3/2}}+\frac{1}{x^4}\left(\frac{xt'}{\sqrt{x^2x'^2+t'^2}}\right)^3=I_1+I_2,\end{aligned} \tag{3.16}$$

where

$$I_1=\frac{x^2\big(\frac{t'}{x'}\big)'x'^2}{\big\{x^2x'^2+t'^2\big\}^{3/2}},\qquad I_2=\frac{1}{x^4}\left(\frac{xt'}{\sqrt{x^2x'^2+t'^2}}\right)^3.$$

From (3.14), taking a derivative with respect to s, we have

$$x'=\frac{-cr\sin(c\tilde{s}-c_1)\sqrt{x^2x'^2+t'^2}}{2x^2}. \tag{3.17}$$

On the other hand, (3.7) implies

$$t'=\big(E-\lambda x^2\big)\frac{\sqrt{x^2x'^2+t'^2}}{x}. \tag{3.18}$$

By means of (3.14), (3.17),(3.18) and (3.1), after direct computations, we have

$$\begin{aligned} I_1 &= \frac{x^2\big(\frac{t'}{x'}\big)'x'^2}{\big\{x^2x'^2+t'^2\big\}^{3/2}} \\ &= \left(\frac{2x\big(E-\lambda x^2\big)}{-cr\sin\left(c\tilde{s}-c_1\right)}\right)'\left(\frac{c^2r^2\sin^2\left(c\tilde{s}-c_1\right)\big(x^2x'^2+t'^2\big)}{4x^4}\right) \\ &= \frac{1}{4x^2}\Bigg(-2x^2x'\big(4\lambda x^2x'-2x'\big(E-\lambda x^2\big)\big)\left(\frac{x^2-\big(E-\lambda x^2\big)^2}{x^4x'^2}\right) \\ &\quad +2\big(E-\lambda x^2\big)\big(c^2x^2-k\big)\Bigg), \end{aligned}$$

and from (3.7), we have

$$I_2 = \frac{1}{x^4}\left(\frac{xt'}{\sqrt{x^2x'^2+t'^2}}\right)^3 = \frac{1}{x^4}\big(E-\lambda x^2\big)^3.$$

Therefore,

$$-2\lambda = I_1+I_2 = \frac{1}{4x^4}\big(\big(-4c-2Ec^2-2c+ck\big)x^4+\big(8\lambda E^2+4E-2Ek\big)x^2\big),$$

which implies $k = 2cE+2$.

3.5. **Horizontal generating curves for $c \neq 0$.** In this subsection, we will show that k, and hence the energy E, has a lower bound. A horizontal generating curve of a rotationally invariant constant p-mean curvature surface is a geodesic curve, which is parametrized by

$$\tilde{\gamma}(\tilde{s}) = \begin{pmatrix} \frac{1}{c}\sin(c\tilde{s})+x_0 \\ -\frac{1}{c}\cos(c\tilde{s})+\frac{1}{c}+y_0 \\ \left(\frac{1}{c^2}+\frac{cy_0}{c^2}\right)\sin(c\tilde{s})+\frac{x_0}{c}\cos(c\tilde{s})-\frac{\tilde{s}}{c}+\frac{\pi}{c^2}-\frac{x_0}{c}+t_0 \end{pmatrix} \tag{3.19}$$

for some (x_0, y_0, t_0), where $\tilde{s}$ is a horizontal arc length parameter.

Suppose that $\gamma(s) = (x(s), 0, t(s))$ with $x \geq 0$ is the corresponding generating curve, we have

$$\begin{aligned} x^2 &= \left(\frac{1}{c}\sin(c\tilde{s})+x_0\right)^2+\left(-\frac{1}{c}\cos(c\tilde{s})+\frac{1}{c}+y_0\right)^2 \\ &= 2\frac{x_0}{c}\sin(c\tilde{s})-2\left(\frac{1+cy_0}{c^2}\right)\cos(c\tilde{s})+x_0^2+\left(\frac{1+cy_0}{c}\right)^2+\frac{1}{c^2} \\ &= r\cos(c\tilde{s}-c_1)+\frac{k}{c^2}, \end{aligned}$$

where

$$\begin{aligned} &k = 1+(cx_0)^2+(1+cy_0)^2 \geq 1, \\ &r = \sqrt{\left(2\frac{x_0}{c}\right)^2+\left(2\left(\frac{1+cy_0}{c^2}\right)\right)^2} = \frac{2}{c^2}\sqrt{(cx_0)^2+(1+cy_0)^2} = \frac{2}{c^2}\sqrt{k-1}, \end{aligned}$$

and c_1 is a real number such that $\sin c_1 = \frac{2\frac{x_0}{c}}{r}$, $\cos c_1 = -\frac{2(\frac{1+cy_0}{c^2})}{r}$.

3.6. **The invariants ζ_1 and ζ_2 for surfaces with $c \neq 0$.** If $r = 0$, then $k = 1$, $x_0 = 0$, $y_0 = -\frac{1}{c}$. Thus, (3.19) implies

$$\tilde{\gamma}(\tilde{s}) = \left(\frac{1}{c}\sin(c\tilde{s}), -\frac{1}{c}\cos(c\tilde{s}), -\frac{\tilde{s}}{c}+\frac{\pi}{c^2}+t_0\right),$$

which generates a cylinder. We assume from now on that $r \neq 0$. Taking the derivative with respect to $\tilde{s}$ on both sides of (3.14) to have $2xx_{\tilde{s}} = -rc\sin(c\tilde{s}-c_1)$. Together with (3.9), we have α of the general form as follows:

$$\alpha = \frac{xx_{\tilde{s}}}{x^2} = \frac{-r\lambda\sin(c\tilde{s}-c_1)}{\frac{k}{c^2}+r\cos(c\tilde{s}-c_1)} = \frac{\lambda\sin(c\tilde{s}-c_1)}{c_2-\cos(c\tilde{s}-c_1)},$$

where $c_2 = -\frac{k}{rc^2}$.

In this subsection, we want to normalize a and b such that they have the forms looking as (2.8) and (2.9), respectively. Together with (3.4), (3.14) and (3.7), we have

$$a = \frac{t'}{x\sqrt{1+\alpha^2}\sqrt{x^2x'^2+t'^2}} = \left(\frac{E}{x^2}-\lambda\right)\frac{1}{\sqrt{1+\alpha^2}},$$
$$b = \frac{1}{x^2\sqrt{1+\alpha^2}} = \frac{1}{\frac{k}{c^2}+r\cos(c\tilde{s}-c_1)}\frac{1}{\sqrt{1+\alpha^2}}.$$

Thus we choose the normal coordinates $\{\bar{s},\bar{\theta}\}$ with $\bar{s}=\tilde{s}+\Gamma(\tilde{\theta})$, $\bar{\theta}=\Psi(\tilde{\theta})$, such that $\Gamma'(\tilde{\theta})=-E$, $\Psi'(\tilde{\theta})=-2\lambda^2 r$. Then we have

$$\bar{a}=\frac{-\lambda}{\sqrt{1+\bar{\alpha}^2}},\qquad \bar{b}=\frac{2\lambda^2}{\left(-\frac{k}{rc^2}-\cos(c\tilde{s}-c_1)\right)}\frac{1}{\sqrt{1+\bar{\alpha}^2}},$$

with

$$\bar{\alpha}=\frac{\lambda\sin(c\tilde{s}-c_1)}{c_2-\cos(c\tilde{s}-c_1)}=\frac{\lambda\sin\left(c\bar{s}-c_1-\frac{E\bar{\theta}}{r\lambda}\right)}{c_2-\cos\left(c\bar{s}-c_1-\frac{E\bar{\theta}}{r\lambda}\right)},$$

that is,

$$\zeta_1(\bar{\theta})=-c_1-\frac{2E\bar{\theta}}{cr},\qquad \zeta_2(\bar{\theta})=c_2=-\frac{k}{rc^2}=-\frac{2cE+2}{c^2r}. \tag{3.20}$$

If $E=0$, then $k=2$, thus the surface has the generating curve defined by

$$x^2=\frac{2}{c^2}+r\cos(c\tilde{s}-c_1)=r\left(\frac{2}{c^2r}+\cos(c\tilde{s}-c_1)\right),$$
$$t=-\lambda\left(\frac{2}{c^2}\tilde{s}+\frac{r}{c}\sin(c\tilde{s}-c_1)\right),$$

with $\zeta_1(\bar{\theta})=-c_1$, $\zeta_2(\bar{\theta})=-\frac{2}{c^2r}<0$. Therefore, we see that $x^2\geq 0\Leftrightarrow\cos(c\tilde{s}-c_1)\geq\zeta_2(\bar{\theta})$, which means that the generating curve (x,t) is defined on the whole $\mathbb{R}$ if and only if $\zeta_2(\bar{\theta})\leq -1$. In particular, if $\zeta_2(\bar{\theta})=-1$, it is the Pansu sphere.

If $E\neq 0$, then $k=2cE+2$ and (3.20) implies that

$$\zeta_1'(\bar{\theta})=-\frac{2E}{cr}>\zeta_2(\bar{\theta}). \tag{3.21}$$

For any constants η_1 and η_2 with $\eta_1>\eta_2$, we obtain the unique solution to the equation system

$$-\frac{2E}{cr}=\eta_1,\qquad -\frac{2cE+2}{c^2r}=\eta_2.$$

3.7. **The allowed values of k and E with $c=0$.** In this subsection, we shall show what possible values can k and E attain. Assume that $c=2\lambda=0$. We write (3.5) as

$$0=\frac{x^3(x't''-x''t')+t'^3}{x\{x^2x'^2+t'^2\}^{3/2}}=\frac{x^2\left(\frac{t'}{x'}\right)'x'^2}{\{x^2x'^2+t'^2\}^{3/2}}+\frac{1}{x^4}\left(\frac{xt'}{\sqrt{x^2x'^2+t'^2}}\right)^3=I_1+I_2.$$

Taking a derivative of (3.15) with respect to s, we get

$$x'=\frac{(k\tilde{s}+k_1)\sqrt{x^2x'^2+t'^2}}{x^2}. \tag{3.22}$$

On the other hand, from (3.7), we have

$$t'=\frac{E\sqrt{x^2x'^2+t'^2}}{x}. \tag{3.23}$$

By means of (3.15), (3.22),(3.23) and (3.1), a direct computation gives

$$I_1=\frac{x^2\left(\frac{t'}{x'}\right)'x'^2}{\{x^2x'^2+t'^2\}^{3/2}}=\frac{1}{x^2}\left(\frac{E(x^2-E^2)}{x^2}-kE\right),$$

and (3.7) implies

$$I_2=\frac{1}{x^4}\left(\frac{xt'}{\sqrt{x^2x'^2+t'^2}}\right)^3=\frac{1}{x^4}E^3.$$

Therefore, $0=I_1+I_2=\frac{E(1-k)}{x^2}$, which says that

$$E=0\qquad\text{or}\qquad k=1. \tag{3.24}$$

The equation (3.7) says that t and $E\tilde{s}$ are differed only by a constant. If $E=0$, then t is constant, which gives us that Σ is a plane that is perpendicular to t-axis.

3.8. **The invariants $\boldsymbol{\zeta_1}$ and $\boldsymbol{\zeta_2}$ for surfaces with $\boldsymbol{c=0}$ and $\boldsymbol{E\neq 0}$.** If $E=0$, the surface is a perpendicular plane to the t-axis. Therefore, in this subsection, we assume that $E\neq 0$, and thus, from (3.24), we have $k=1$. Then one rewrites α to be

$$\alpha=\frac{xx_{\tilde{s}}}{x^2}=\frac{\tilde{s}+k_1}{(\tilde{s}+k_1)^2+\big(k_2-k_1^2\big)}.$$

From (3.15), we have $\frac{1}{x^2}=\frac{\alpha}{\tilde{s}+k_1}>0$. We want to normalize a and b such that they have the form specified in [5, Theorem 1.3]. Together with (3.4) and (3.7), we have

$$\begin{aligned}a&=\frac{t'}{x\sqrt{1+\alpha^2}\sqrt{x^2x'^2+t'^2}}=\left(\frac{E}{x^2}\right)\frac{1}{\sqrt{1+\alpha^2}}=Eb,\\ b&=\frac{1}{x^2\sqrt{1+\alpha^2}}=\frac{\alpha}{(\tilde{s}+k_1)\sqrt{1+\alpha^2}}=\frac{|\alpha|}{(|\tilde{s}+k_1|)\sqrt{1+\alpha^2}}.\end{aligned}$$

Thus we choose the normal coordinates $\{\bar{s},\bar{\theta}\}$ with $\bar{s}=\tilde{s}+\Gamma\big(\tilde{\theta}\big)$, $\bar{\theta}=\Psi\big(\tilde{\theta}\big)$ such that $\Gamma'\big(\tilde{\theta}\big)=-E$, $\Psi'\big(\tilde{\theta}\big)=1$. Then we have

$$\bar{a}=0,\qquad \bar{b}=\frac{|\bar{\alpha}|}{|\tilde{s}+k_1|\sqrt{1+\bar{\alpha}^2}}$$

with

$$\bar{\alpha}=\frac{\tilde{s}+k_1}{(\tilde{s}+k_1)^2+\big(k_2-k_1^2\big)}=\frac{\bar{s}-\Gamma\big(\tilde{\theta}\big)+k_1}{(\bar{s}-\Gamma\big(\tilde{\theta}\big)+k_1)^2+\big(k_2-k_1^2\big)},$$

that is,

$$\begin{aligned}&\zeta_1\big(\bar{\theta}\big)=k_1+E\bar{\theta},\qquad \text{which is linear in } \bar{\theta},\\ &\zeta_2\big(\bar{\theta}\big)=k_2-k_1^2,\qquad \text{which is a constant, denoted as } \zeta_2.\end{aligned}$$

From (3.15) and (3.7), we conclude that the generating curve is defined by

$$x^2=(\tilde{s}+k_1)^2+\zeta_2,\qquad t=E\tilde{s},\qquad \text{up to a constant.}\tag{3.25}$$

Remark 3.3. We remark that for $\lambda=0$, two kinds of p-minimal surfaces are presented depending on the energy E. When $E=0$, t in (3.16) is constant and then one obtains a plane that is perpendicular to the t-axis. On the other hand, if $E\neq 0$, we have p-minimal surfaces generated by curves defined by (3.25). For $\lambda\neq 0$, substituting (3.14) in (3.8), we see

$$\begin{aligned}t&=E\tilde{s}-\lambda\int\left(\frac{k}{c^2}+\sqrt{k_1^2+k_2^2}\cos(c\tilde{s}-c_1)\right)\mathrm{d}\tilde{s}\\ &=\left(E-\frac{k}{4\lambda}\right)\tilde{s}-\frac{\sqrt{k_1^2+k_2^2}}{2}\sin(2\lambda\tilde{s}-c_1)+\text{const.}\end{aligned}$$

In the case $\lambda\neq 0$, we give the following two examples.

Example 3.4. We choose k_1, k_2 in (3.13) so that $\sqrt{k_1^2+k_2^2}=-\frac{k}{c^2}$, and then (3.14) implies

$$x=\pm\frac{\sqrt{2k}}{2\lambda}\sin\left(\lambda\tilde{s}-\frac{c_1}{2}\right).$$

Moreover, if $E=0$, then $t=-\frac{k}{4\lambda}\tilde{s}+\frac{k}{8\lambda^2}\sin(2\lambda\tilde{s}-c_1)$, which is a scaling sphere.

The other two integrability conditions (see [5, equation (2.13)]) are

$$-\frac{b_{\tilde{s}}}{b}=2\alpha+\frac{\alpha\alpha_{\tilde{s}}}{1+\alpha^2},\qquad aH_{\tilde{s}}+bH_{\tilde{\theta}}=\frac{\alpha_{\tilde{s}\tilde{s}}+6\alpha\alpha_{\tilde{s}}+4\alpha^3+\alpha H^2}{\sqrt{1+\alpha^2}}.\tag{3.26}$$

We rewrite the first equation in (3.26) as

$$2\alpha+\frac{\alpha\alpha_{\tilde{s}}}{1+\alpha^2}+\frac{b_{\tilde{s}}}{b}=0.$$

Integrating on both sides to see that

$$\int\left(2\alpha+\frac{\alpha\alpha_{\tilde{s}}}{1+\alpha^2}+\frac{b_{\tilde{s}}}{b}\right)\mathrm{d}\tilde{s}$$

is a constant. More precisely, in terms of x, x', t, t', we write

$$\int 2\left(\alpha+\frac{\alpha\alpha_{\tilde{s}}}{1+\alpha^2}+\frac{b_{\tilde{s}}}{b}\right)\mathrm{d}\tilde{s}=\int\left(2\frac{x_{\tilde{s}}}{x}+\frac{\alpha\alpha_{\tilde{s}}}{1+\alpha^2}+\frac{b_{\tilde{s}}}{b}\right)\mathrm{d}\tilde{s}=\ln\big(bx^2\sqrt{1+\alpha^2}\big)+\text{const.}$$

The conclusion is that $\ln\big(bx^2\sqrt{1+\alpha^2}\big)$ is a constant, which also follows from (3.4).

Suppose H is constant. The second equation of (3.26) is exactly $\alpha_{\tilde{s}\tilde{s}}+6\alpha\alpha_{\tilde{s}}+4\alpha^3+\alpha H^2=0$. Using (3.9), this ODE becomes (3.10), which has been discussed previously.

4. The construction of constant p-mean curvature surfaces

In this section, we construct constant p-mean curvature surfaces by perturbing the Pansu sphere in some way. Recall the parametrization of the Pansu sphere (2.3). For each fixed angle θ, the curve l_θ defined by $l_\theta(s) = (x(s)\cos\theta - y(s)\sin\theta, x(s)\sin\theta + y(s)\cos\theta, t(s))$ is a geodesic with curvature 2λ. Let $\mathcal{C}$ be an arbitrary curve $\mathcal{C}\colon \mathbb{R} \to H_1$ given by $\mathcal{C}(\theta) = (x_1(\theta), x_2(\theta), x_3(\theta))$. For each fixed θ, we translate l_θ by $\mathcal{C}(\theta)$, so that the curve $L_{\mathcal{C}(\theta)}(l_\theta)$ is also a geodesic curve with curvature 2λ. Then the union of all these curves $\Sigma_\mathcal{C} = \cup_\theta L_{\mathcal{C}(\theta)}(l_\theta)$ constitutes *a constant p-mean curvature surface* with a parametrization

$$\begin{aligned} Y(s,\theta) = (&x_1(\theta) + \big(x(s)\cos\theta - y(s)\sin\theta\big), x_2(\theta) + (x(s)\sin\theta + y(s)\cos\theta),\\ &x_3(\theta) + t(s) + x_2(\theta)(x(s)\cos\theta - y(s)\sin\theta)\\ &- x_1(\theta)(x(s)\sin\theta + y(s)\cos\theta)). \end{aligned} \tag{4.1}$$

By a straightforward computation, and notice that

$$x'(s)\cos\theta - y'(s)\sin\theta = \cos(2\lambda s+\theta), \qquad x'(s)\sin\theta + y'(s)\cos\theta = \sin(2\lambda s+\theta),$$
$$x(s)\cos\theta - y(s)\sin\theta = -\frac{1}{2\lambda}(\sin\theta - \sin(2\lambda s+\theta)),$$
$$x(s)\sin\theta + y(s)\cos\theta = \frac{1}{2\lambda}(\cos\theta - \cos(2\lambda s+\theta)), \qquad x^2(s)+y^2(s) = \frac{1}{2\lambda^2}(1-\cos 2\lambda s),$$

we have

$$\begin{aligned} Y_s &= \big(x'(s)\cos\theta - y'(s)\sin\theta\big)\mathring{e}_1 + \big(x'(s)\sin\theta + y'(s)\cos\theta\big)\mathring{e}_2|_{Y(s,\theta)}\\ &= \cos(2\lambda s+\theta)\mathring{e}_1 + \sin(2\lambda s+\theta)\mathring{e}_2,\\ Y_\theta &= \big(x_1'(\theta) - x(s)\sin\theta - y(s)\cos\theta\big)\mathring{e}_1 + \big(x_2'(\theta) + x(s)\cos\theta - y(s)\sin\theta\big)\mathring{e}_2 + \big(\Theta(\mathcal{C}'(\theta))\\ &\quad + 2x_2'(\theta)\big(x(s)\cos\theta - y(s)\sin\theta\big) - 2x_1'(\theta)\big(x(s)\sin\theta + y(s)\cos\theta\big) + x^2(s)+y^2(s)\big)T\\ &= \left(x_1'(\theta) - \frac{1}{2\lambda}(\cos\theta - \cos(2\lambda s+\theta))\right)\mathring{e}_1 + \left(x_2'(\theta) - \frac{1}{2\lambda}(\sin\theta - \sin(2\lambda s+\theta))\right)\mathring{e}_2\\ &\quad + \left(\Theta(\mathcal{C}'(\theta)) - x_2'(\theta)\frac{1}{\lambda}(\sin\theta - \sin(2\lambda s+\theta)) - x_1'(\theta)\frac{1}{\lambda}(\cos\theta - \cos(2\lambda s+\theta))\right.\\ &\quad \left. + \frac{1}{2\lambda^2}(1-\cos 2\lambda s)\right)T. \end{aligned}$$

Therefore,

$$\begin{aligned} Y_s \wedge Y_\theta = &\left[x_2'(\theta)\cos(2\lambda s+\theta) - x_1'(\theta)\sin(2\lambda s+\theta) + \frac{\sin 2\lambda s}{2\lambda}\right]\mathring{e}_1\wedge\mathring{e}_2\\ &+ [\cos(2\lambda s+\theta)\langle Y_\theta, T\rangle]\mathring{e}_1\wedge T + [\sin(2\lambda s+\theta)\langle Y_\theta, T\rangle]\mathring{e}_2\wedge T\\ = &\left[A(\theta)\cos 2\lambda s + \left(\frac{1}{2\lambda} - B(\theta)\right)\sin 2\lambda s\right]\mathring{e}_1\wedge\mathring{e}_2\\ &+ [\cos(2\lambda s+\theta)\langle Y_\theta, T\rangle]\mathring{e}_1\wedge T + [\sin(2\lambda s+\theta)\langle Y_\theta, T\rangle]\mathring{e}_2\wedge T, \end{aligned} \tag{4.2}$$

where

$$A(\theta) = x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta, \qquad B(\theta) = x_2'(\theta)\sin\theta + x_1'(\theta)\cos\theta,$$
$$\langle Y_\theta, T\rangle = \frac{1}{\lambda}\left[\left(B(\theta) - \frac{1}{2\lambda}\right)\cos 2\lambda s + A(\theta)\sin 2\lambda s + D(\theta)\right],$$
$$D(\theta) = \lambda\Theta(\mathcal{C}'(\theta)) + \left(\frac{1}{2\lambda} - B(\theta)\right). \tag{4.3}$$

From (4.2), we conclude that Y is *an immersion* if and only if either

$$\left[A(\theta)\cos 2\lambda s + \left(\frac{1}{2\lambda} - B(\theta)\right)\sin 2\lambda s\right] \neq 0 \qquad \text{or} \qquad \langle Y_\theta, T\rangle \neq 0.$$

For the constructed surface Y in (4.1), we always assume it is defined on a region such that Y is an immersion and $\Sigma_\mathcal{C}$ is the constant p-mean curvature surface defined by such an immersion Y. A point $p \in \Sigma_\mathcal{C}$ is *a singular point* if and only if $\langle Y_\theta, T\rangle = 0$. Thus at a singular point, we must have

$$\left[A(\theta)\cos 2\lambda s + \left(\frac{1}{2\lambda} - B(\theta)\right)\sin 2\lambda s\right] \neq 0.$$

Now, we proceed to compute the invariants for Y. From the construction of Y, we see that (s,θ) is a compatible coordinate system and we are able to choose the characteristic direction $e_1 = Y_s$, and hence

$$e_2 = Je_1 = -\sin(2\lambda s+\theta)\mathring{e}_1 + \cos(2\lambda s+\theta)\mathring{e}_2.$$

The α-function is a function defined on the regular part that satisfies

$$\alpha e_2 + T = a\sqrt{1+\alpha^2}Y_s + b\sqrt{1+\alpha^2}Y_\theta = a\sqrt{1+\alpha^2}e_1 + b\sqrt{1+\alpha^2}Y_\theta$$

for some functions a and b. This is equivalent to, comparing the alike terms,

$$\begin{aligned}
-\alpha\sin(2\lambda s+\theta) &= a\sqrt{1+\alpha^2}\cos(2\lambda s+\theta)\\
&\quad + b\sqrt{1+\alpha^2}\left(x_1'(\theta) - \frac{1}{2\lambda}(\cos\theta - \cos(2\lambda s+\theta))\right),\\
\alpha\cos(2\lambda s+\theta) &= a\sqrt{1+\alpha^2}\sin(2\lambda s+\theta)\\
&\quad + b\sqrt{1+\alpha^2}\left(x_2'(\theta) - \frac{1}{2\lambda}(\sin\theta - \sin(2\lambda s+\theta))\right),\\
1 &= b\sqrt{1+\alpha^2}\langle Y_\theta, T\rangle.
\end{aligned}$$

We thus have

$$\begin{aligned}
a &= \frac{-2\lambda\big(x_1'(\theta)\cos(2\lambda s+\theta) + x_2'(\theta)\sin(2\lambda s+\theta)\big) - (1-\cos 2\lambda s)}{2\lambda\sqrt{1+\alpha^2}\langle Y_\theta, T\rangle},\\
b &= \frac{1}{\sqrt{1+\alpha^2}\langle Y_\theta, T\rangle},\\
\alpha &= \frac{x_2'(\theta)\cos(2\lambda s+\theta) - x_1'(\theta)\sin(2\lambda s+\theta) + \frac{\sin 2\lambda s}{2\lambda}}{\langle Y_\theta, T\rangle}\\
&= \lambda\frac{A(\theta)\cos 2\lambda s + \big(\frac{1}{2\lambda} - B(\theta)\big)\sin 2\lambda s}{\big(B(\theta) - \frac{1}{2\lambda}\big)\cos 2\lambda s + A(\theta)\sin 2\lambda s + D(\theta)}.
\end{aligned} \tag{4.4}$$

Let $V = \big(A(\theta), \frac{1}{2\lambda} - B(\theta)\big)$ and $\|V\| = \sqrt{[A(\theta)]^2 + \big[\frac{1}{2\lambda} - B(\theta)\big]^2}$. If $V = 0$, then $\alpha = 0$. If $V \neq 0$, then we can write $\frac{V}{\|V\|} = (\sin\zeta(\theta), \cos\zeta(\theta))$, for some function $\zeta(\theta)$. The functions α, a, and b can be further written as

$$\begin{aligned}
\alpha &= \lambda\frac{\sin\zeta(\theta)\cos 2\lambda s + \cos\zeta(\theta)\sin 2\lambda s}{-\cos\zeta(\theta)\cos 2\lambda s + \sin\zeta(\theta)\sin 2\lambda s + \frac{D(\theta)}{\|V\|}} = \lambda\frac{\sin\big(2\lambda s+\zeta(\theta)\big)}{G(\theta) - \cos\big(2\lambda s+\zeta(\theta)\big)},\\
a &= \frac{-2\lambda[\sin\zeta(\theta)\sin(2\lambda s) - \cos\zeta(\theta)\cos 2\lambda s] - \frac{1}{\|V\|}}{2\sqrt{1+\alpha^2}\big[\frac{D(\theta)}{\|V\|} - \cos(2\lambda s+\zeta(\theta))\big]}\\
&= \frac{2\lambda\|V\|\cos(2\lambda s+\zeta(\theta)) - 1}{2\|V\|\sqrt{1+\alpha^2}[G(\theta) - \cos(2\lambda s+\zeta(\theta))]},\\
b &= \frac{\frac{\lambda}{\|V\|}}{\sqrt{1+\alpha^2}\big[\frac{D(\theta)}{\|V\|} - \cos(2\lambda s+\zeta(\theta))\big]} = \frac{\frac{\lambda}{\|V\|}}{\sqrt{1+\alpha^2}[G(\theta) - \cos(2\lambda s+\zeta(\theta))]},
\end{aligned}$$

where

$$G(\theta) = \frac{D(\theta)}{\|V\|} = \frac{D(\theta)}{\sqrt{(A(\theta))^2 + \big(\frac{1}{2\lambda} - B(\theta)\big)^2}}. \tag{4.5}$$

Next, we normalize the three invariants α, a, and b. Firstly, we choose another compatible coordinates $(\tilde{s} = s + \Gamma(\theta),\ \tilde{\theta} = \Psi(\theta))$, for some $\Gamma(\theta)$ and $\Psi(\theta)$. From the transformation law of the induced metric $\tilde{a} = a + b\Gamma'(\theta)$, $\tilde{b} = b\Psi'(\theta)$, this can be chosen so that

$$\Gamma'(\theta) = \frac{-2\lambda\|V(\theta)\|G(\theta) + 1}{2\lambda} = \frac{1}{2\lambda} - D(\theta),$$

or equivalently,

$$\Gamma(\theta) = \frac{\theta}{2\lambda} - \int D(\theta)\mathrm{d}\theta.$$

If we further choose Ψ such that $\tilde{\theta} = \Psi(\theta) = 2\lambda \int \|V(\theta)\| \mathrm{d}\theta$, then in terms of the compatible coordinates $(\tilde{s}, \tilde{\theta})$, the three invariants read

$$\tilde{a} = \frac{-\lambda}{\sqrt{1+\tilde{\alpha}^2}}, \qquad \tilde{b} = \frac{2\lambda^2}{\sqrt{1+\tilde{\alpha}^2}\{G(\Psi^{-1}(\tilde{\theta})) - \cos[2\lambda\tilde{s} - 2\lambda\Gamma(\Psi^{-1}(\tilde{\theta})) + \zeta(\Psi^{-1}(\tilde{\theta}))]\}},$$
$$\tilde{\alpha} = \lambda \frac{\sin(2\lambda\tilde{s} - 2\lambda\Gamma(\theta) + \zeta(\Psi^{-1}(\tilde{\theta})))}{G(\Psi^{-1}(\tilde{\theta})) - \cos[2\lambda\tilde{s} - 2\lambda\Gamma(\Psi^{-1}(\tilde{\theta})) + \zeta(\Psi^{-1}(\tilde{\theta}))]},$$

where D and G are defined in (4.3) and (4.5), respectively.

We summarize the above discussion as a theorem in the following.

Theorem 4.1. *The coordinate system (s, θ) for Y in (4.1) is compatible. If $V = 0$, then $\alpha = 0$. If $V \neq 0$, then the new coordinate system $(\tilde{s}, \tilde{\theta})$, where $\tilde{s} = s + \Gamma(\theta)$, $\tilde{\theta} = \Psi(\theta)$, with*

$$\Gamma(\theta) = \frac{\theta}{2\lambda} - \int D(\theta)\mathrm{d}\theta, \qquad \Psi(\theta) = 2\lambda \int \|V(\theta)\|\mathrm{d}\theta,$$

is normal. In terms of the normal coordinates, the invariants of Y are given by

$$\zeta_1(\tilde{\theta}) = \zeta(\Psi^{-1}(\tilde{\theta})) - 2\lambda\Gamma(\Psi^{-1}(\tilde{\theta})), \qquad \zeta_2(\tilde{\theta}) = G(\Psi^{-1}(\tilde{\theta})). \tag{4.6}$$

Particularly, in order to have constant $\zeta_1(\tilde{\theta})$ and nonzero constant $\zeta_2(\tilde{\theta})$, Theorem 4.1 suggests the constant p-mean curvature surfaces deformed by curves

$$\mathcal{C}(\theta) = (x_1(\theta), x_2(\theta), x_3(\theta)) = \left(\frac{r}{\lambda}\sin\theta, -\frac{r}{\lambda}\cos\theta, \frac{r(1-r)}{\lambda^2}\theta\right), \tag{4.7}$$

where $r \neq \frac{1}{2}$. More precisely, we have the following proposition.

Proposition 4.2. *For any curve $\mathcal{C}(\theta)$ defined as (4.7), the deformed surface $Y(s, \theta)$ has both constant invariants $\zeta_1(\tilde{\theta})$ and $\zeta_2(\tilde{\theta}) \neq 0$.*

Proof. We argue by assuming $\zeta_2(\tilde{\theta}) = \zeta_2$ is a constant, $x_1(\theta) = \frac{r}{\lambda}\sin\theta$, and $x_2(\theta) = -\frac{r}{\lambda}\cos\theta$ for any $r \neq \frac{1}{2}$. Then (4.3) implies $A(\theta) = 0$, $B(\theta) = \frac{r}{\lambda}$, which leads to $\|V\| = \frac{|1-2r|}{2\lambda}$. The second equation of (4.6) shows that $D(\theta) = \zeta_2\|V\|$, and hence

$$\begin{aligned}\zeta_1(\tilde{\theta}) &= \sin^{-1}\left(\frac{A(\theta)}{\|V\|}\right) - 2\lambda\left(\frac{\theta}{2\lambda} - \int D(\theta)\mathrm{d}\theta\right) = -\theta + \int \zeta_2|1-2r|\mathrm{d}\theta \\ &= (\zeta_2|1-2r| - 1)\theta + \text{const}.\end{aligned}$$

In order to have $\zeta_1(\tilde{\theta})$ being constant, we must have $\zeta_2|1-2r| = 1$. It is clear to see that $\zeta_2 \neq 0$ and $r \neq 0$. The system (4.6) immediately shows $D(\theta) = \frac{1}{2\lambda}$, which gives $x_3'(\theta) = \frac{r(1-r)}{\lambda^2}$, by (4.3). Namely, $x_3(\theta) = \frac{r(1-r)}{\lambda^2}\theta + \text{const}$.

Moreover, the new coordinates can be obtained by

$$\tilde{\theta} = \Psi(\theta) = |1-2r|\theta + \text{const} \qquad \text{and} \qquad \Gamma(\theta),$$

up to a constant. □

5. Examples

It is easy to see that the Pansu sphere can be obtained by deforming the following curves

$$\mathcal{C}_1(\theta) = (0, 0, \text{const}) \qquad \text{or} \qquad \mathcal{C}_2(\theta) = \left(\frac{1}{\lambda}\sin\theta, -\frac{1}{\lambda}\cos\theta, \text{const}\right).$$

Using a similar idea as Theorem 4.1 and Proposition 4.2, we obtain curves $\mathcal{C}(\theta)$ that result in constant p-mean curvature surfaces with constant ζ_2 and linear $\zeta_1(\tilde{\theta})$ in Sections 5.1 and 5.3. We collect $\mathcal{C}(\theta)$ in Tables 1 and 2 as follows.

5.1. Examples of constant p-mean curvature surfaces.

Proposition 5.1. *Given any curve*

$$\mathcal{C}(\theta) = \left(\frac{1}{\lambda}\sin\theta, -\frac{1}{\lambda}\cos\theta, \frac{k-1}{2\lambda^2}\theta + \text{const}\right),$$

the deformed surface $Y(s, \theta)$ has the invariants $\zeta_1(\tilde{\theta}) = (k-1)\tilde{\theta} + \text{const}$ and $\zeta_2(\tilde{\theta}) = k$, where $k \in \mathbb{R}$.

Remark 5.2. It is easy to see that the surfaces obtained by curves given in Proposition 5.1 are not rotationally symmetric since $\zeta_1' = k - 1 < \zeta_2$ by (3.21).

Proposition 5.3. *For any constant $k > 0$ and m, there exist constant p-mean curvature surfaces $Y(s, \theta)$ defined as (4.1) with invariants $\zeta_1(\theta) = m\theta + \text{const}$ and $\zeta_2 = k$.*

TABLE 1. Examples of $\mathcal{C}(\theta)$ for constant p-mean curvature surfaces.

$\mathcal{C}(\theta)$	constant ζ_1	linear ζ_1
$\zeta_2 > 1$	$\big(\frac{r}{\lambda}\sin\theta, -\frac{r}{\lambda}\cos\theta, \frac{r(1-r)}{\lambda^2}\theta\big)$ $0 < r < 1, r \neq \frac{1}{2}$ $\zeta_2 = \frac{1}{\lvert 1-2r\rvert}$	$m \neq 1$: $(x_1(\theta), x_2(\theta), x_3(\theta))$ $x_1(\theta) = \frac{\sin\theta}{2\lambda} - \frac{\sin((m-1)\theta)}{2\lambda k(m-1)}$ $x_2(\theta) = -\frac{\cos\theta}{2\lambda} - \frac{\cos((m-1)\theta)}{2\lambda k(m-1)}$ $x_3(\theta) = \frac{1+k^2(m-1)}{4\lambda^2 k^2(m-1)}\theta - \frac{\sin(m\theta)}{4\lambda^2 k(m-1)}$ $m = 1$: $\big(\frac{\sin\theta}{2\lambda} - \frac{\theta}{2\lambda k}, -\frac{\cos\theta}{2\lambda}, \frac{k\theta - \theta\cos\theta}{4\lambda^2 k}\big)$ $\zeta_1 = m\theta + \text{const}$ and $\zeta_2 = k > 0$
$\zeta_2 = 1$	Pansu sphere	
$0 < \zeta_2 < 1$	$\big(\frac{r}{\lambda}\sin\theta, -\frac{r}{\lambda}\cos\theta, \frac{r(1-r)}{\lambda^2}\theta\big)$ $r < 0$ or $r > 1$ $\zeta_2 = \frac{1}{\lvert 1-2r\rvert}$	
$\zeta_2 = 0$	$\big(\frac{\beta}{4\lambda}, 0, \frac{\beta}{4\lambda} - \frac{\theta}{4\lambda^2}\big)$ $\beta = \ln\lvert\sec\theta + \tan\theta\rvert$ $\zeta_1 = 0$	$\big(\cos\theta, \sin\theta, -\big(\theta + \frac{\theta}{2\lambda^2}\big)\big)$ $\zeta_1 = -\theta + \text{const}$

TABLE 2. Examples of $\mathcal{C}(\theta)$ for p-minimal surfaces.

$\mathcal{C}(\theta)$	constant ζ_1	linear ζ_1
$\zeta_2 > 0$	type I	$(r\sin\theta, -r\cos\theta, z(\theta))$ $z'(\theta) + r^2 > 0$ $\zeta_1 = -r\theta$
$\zeta_2 < 0$	type II, III	$(r\sin\theta, -r\cos\theta, z(\theta))$ $z'(\theta) + r^2 < 0$ $\zeta_1 = -r\theta$
special type I	degenerate case: $\big(-\theta, 0, \frac{\sin(2\theta)-2\theta}{4}\big)$ or entire graph: $u = 0$	
special type II	$u = xy + g(y)$	

Proof. It suffices to solve the system (4.6). In order to obtain a surface with linear $\zeta_1(\theta) = m\theta$ for any given nonzero constant $\zeta_2 = k$, we assume

$$A(\theta) = \frac{1}{2\lambda k}\sin(m\theta) \qquad \text{and} \qquad \frac{1}{2\lambda} - B(\theta) = \frac{1}{2\lambda k}\cos(m\theta). \tag{5.1}$$

It results in $\|V\| = \frac{1}{2\lambda k}$, and

$$\begin{aligned}\zeta_1(\theta) &= \sin^{-1}\left(\frac{A(\theta)}{\|V\|}\right) - 2\lambda\left(\frac{\theta}{2\lambda} - \int D(\theta)\,\mathrm{d}\theta\right) = m\theta - \theta + 2\lambda\int \zeta_2(\theta)\|V\|\,\mathrm{d}\theta \\ &= m\theta - \theta + 2\lambda k\int \frac{1}{2\lambda k}\,\mathrm{d}\theta = m\theta + \text{const}.\end{aligned}$$

Next we solve for $x_1'(\theta)$ and $x_2'(\theta)$ from (4.3), that is,

$$\frac{1}{2\lambda k}\sin(m\theta) = x_2'(\theta)\cos\theta - x_1'(\theta)\sin\theta, \qquad \frac{1}{2\lambda} - \frac{1}{2\lambda k}\cos(m\theta) = x_2'(\theta)\sin\theta + x_1'(\theta)\cos\theta.$$

It is easy to see that

$$x_1'(\theta) = \frac{1}{2\lambda}\cos(\theta) - \frac{1}{2\lambda k}\cos((m-1)\theta), \qquad x_2'(\theta) = \frac{1}{2\lambda}\sin(\theta) + \frac{1}{2\lambda k}\sin((m-1)\theta),$$

and hence for $m \neq 1$,

$$\begin{aligned}x_1(\theta) &= \frac{1}{2\lambda}\sin(\theta) - \frac{1}{2\lambda k(m-1)}\sin((m-1)\theta) + \text{const}, \\ x_2(\theta) &= -\frac{1}{2\lambda}\cos(\theta) - \frac{1}{2\lambda k(m-1)}\cos((m-1)\theta) + \text{const}.\end{aligned} \tag{5.2}$$

The equation (4.3) also suggests

$$x_3'(\theta) = \frac{1+k^2(m-1)}{4\lambda^2 k^2(m-1)} - \frac{m\cos(m\theta)}{4\lambda^2 k(m-1)},$$

and then we have

$$x_3(\theta) = \frac{1+k^2(m-1)}{4\lambda^2 k^2(m-1)}\theta - \frac{\sin(m\theta)}{4\lambda^2 k(m-1)} + \text{const.} \tag{5.3}$$

Therefore, deforming such curves $\mathcal{C}(\theta) = (x_1(\theta), x_2(\theta), x_3(\theta))$ defined by (5.2) and (5.3) gives surfaces with nonzero $\zeta_2 = k$ and linear $\zeta_1(\theta) = m\theta + \text{const}$ for all $m \neq 1$.

When $m = 1$, direct computations from (5.1) imply

$$x_1(\theta) = \frac{1}{2\lambda}\sin(\theta) - \frac{\theta}{2\lambda k} + \text{const}, \qquad x_2(\theta) = -\frac{1}{2\lambda}\cos(\theta) + \text{const},$$
$$x_3(\theta) = \frac{k\theta - \theta\cos\theta}{4\lambda^2 k}.$$

□

Example 5.4. If

$$\begin{aligned} C(\theta) &= (x_1(\theta), x_2(\theta), x_3(\theta)) \\ &= \left(\frac{1}{4\lambda}\ln|\sec\theta + \tan\theta| + c_3, c_4, \frac{c_5}{4\lambda}\ln|\sec\theta + \tan\theta| - \frac{1}{4\lambda^2}\theta + c_6\right), \end{aligned}$$

then $\zeta_1(\tilde{\theta}) = 0$, $\zeta_2(\tilde{\theta}) = 0$.

5.2. **Basic properties of surfaces of special type I.** For p-minimal surfaces of special type I, we have the first fundamental form, in terms of normal coordinates (x, y),

$$I = \mathrm{d}x \otimes \mathrm{d}x + \left(\frac{1+\alpha^2}{\alpha^4}\right)\mathrm{d}y \otimes \mathrm{d}y,$$

so that I degenerates along the curve where α blows up. Recall that the parametrization of the surface Y is

$$Y(r,\theta) = (x(\theta) + r\cos\theta, y(\theta) + r\sin\theta, z(\theta) + ry(\theta)\cos\theta - rx(\theta)\sin\theta).$$

We have

$$Y_r = (\cos\theta, \sin\theta, y(\theta)\cos\theta - x(\theta)\sin\theta), \qquad Y_\theta = \big(x'(\theta) - r\sin\theta, y'(\theta) + r\cos\theta, *\big),$$

where

$$* = z'(\theta) + ry'(\theta)\cos\theta - y(\theta)\sin\theta - x'(\theta)\sin\theta - x(\theta)\cos\theta.$$

Then

$$\begin{aligned} Y_r \times Y_\theta &= \begin{vmatrix} i & j & k \\ \cos\theta & \sin\theta & y(\theta)\cos\theta - x(\theta)\sin\theta \\ x'(\theta) - r\sin\theta & y'(\theta) + r\cos\theta & * \end{vmatrix} \\ &= \rho\big(\sin\theta\big(y'(\theta)\cos\theta - x'(\theta)\sin\theta\big) - y(\theta), \\ &\quad -\cos\theta\big(y'(\theta)\cos\theta - x'(\theta)\sin\theta\big) + x(\theta), 1\big), \end{aligned}$$

where $\rho = r + \big(y'(\theta)\cos\theta - x'(\theta)\sin\theta\big)$. For p-minimal surfaces of special type I, we have

$$\alpha = \frac{1}{r + \big(y'(\theta)\cos\theta - x'(\theta)\sin\theta\big)}.$$

Therefore, Y_r and Y_θ are linearly dependent along the curve when α blows up.

For constant p-mean curvature surfaces of special type I, (4.2) and (4.3) immediately imply that $Y_s \wedge Y_\theta = 0$ if and only if

$$\begin{aligned} 0 &= A(\theta)\cos 2\lambda s + \left(\frac{1}{2\lambda} - B(\theta)\right)\sin 2\lambda s, \\ -D(\theta) &= -\left(\frac{1}{2\lambda} - B(\theta)\right)\cos 2\lambda s + A(\theta)\sin 2\lambda s, \end{aligned}$$

that is,

$$\begin{pmatrix} \cos 2\lambda s & \sin 2\lambda s \\ \sin 2\lambda s & -\cos 2\lambda s \end{pmatrix}\begin{pmatrix} A(\theta)/\|V\| \\ \big(\frac{1}{2\lambda} - B(\theta)\big)/\|V\| \end{pmatrix} = \begin{pmatrix} 0 \\ -G(\theta) \end{pmatrix}.$$

This implies that $Y_s \wedge Y_\theta = 0$ holds if and only if $G(\theta) = \pm 1$, namely, it happens only on the surface Y of special type I at points where the function α blows up.

5.3. **Examples of p-minimal surfaces.** In what follows, we give some p-minimal surfaces of special type II (i.e., $\zeta_2 < 0$ and linear ζ_1). We first recall in [5] that $\mathcal{C}(\theta) = (x(\theta), y(\theta), z(\theta))$ satisfying

$$\zeta_1(\theta) = -\Gamma(\theta) + y'(\theta)\cos\theta - x'\sin\theta,$$
$$\zeta_2(\theta) = z'(\theta) + x(\theta)y'(\theta) - y(\theta)x'(\theta) - \big(y'(\theta)\cos\theta - x'(\theta)\sin\theta\big)^2,$$

where $\Gamma(\theta) = \int x'(\theta)\cos\theta + y'(\theta)\sin\theta \mathrm{d}\theta$, will result in a p-minimal surface. For any nonzero $r \in \mathbb{R}$, if $x(\theta) = r\sin\theta$ and $y(\theta) = -r\cos\theta$, then $\Gamma(\theta) = r\theta$ (up to a constant), $\zeta_1(\theta) = -r\theta$, and $\zeta_2(\theta) = z'(\theta) + r^2$. We choose $z(\theta)$ such that $z'(\theta) + r^2 < 0$ to have negative ζ_2.

Acknowledgements. The first author's research was supported in part by NSTC 112-2115-M-007-009-MY3. The second author's research was supported in part by NSTC 110-2115-M-167-002-MY2 and NSTC 112-2115-M-167-002-MY2. The third author's research was supported in part by NSTC 112-2628-M-032-001-MY4. We all thank the anonymous referees for carefully reading our manuscript and their insightful comments and suggestions for improving the article.

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